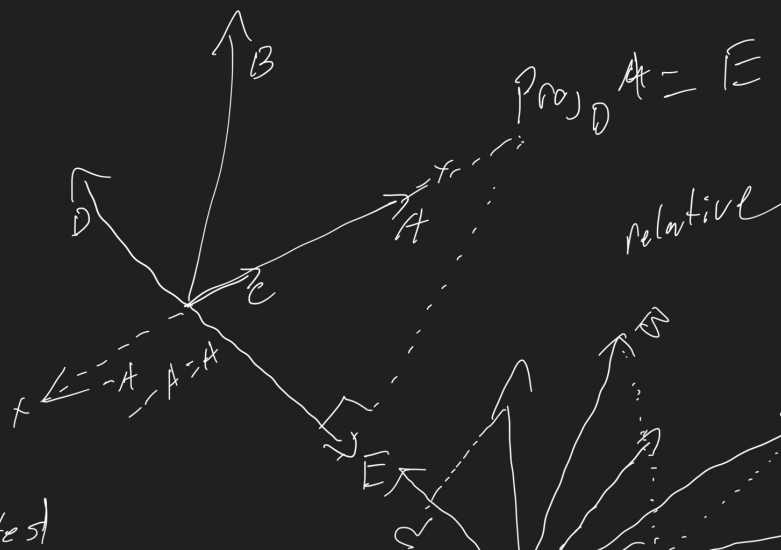
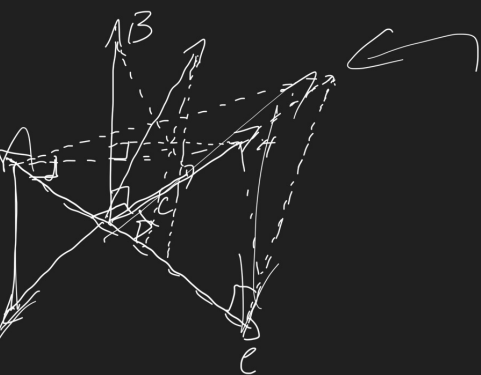


Quiz 1 → linear combinations
 • dot product
 • projection



add more vector operation notes

Cross Product

• special vector product, or cross product in $\mathbb{R}^3$ only.

$$v \times w = u$$

"A vector $\times$ vector is a vector."

• bilinearity:

$$u \times (cv + dw) = c(u \times v) + d(u \times w)$$

• skew symmetry

$$v \times w = -w \times v$$

$$\vec{v} \times \vec{v} = -(\vec{v} \times \vec{v}) = \vec{0}$$

A vector crossed with itself is 0.

• basis vectors

$$i \times j = k,$$

$$j \times k = i,$$

$$k \times i = j$$

ex:

$$\vec{w} = (a\vec{i} + b\vec{j} + c\vec{k}) \times (d\vec{i} + e\vec{j} + f\vec{k})$$

$\hat{i}$	$\hat{j}$	$\hat{k}$
a	b	c
d	e	f

$$\vec{w} = \begin{bmatrix} b \cdot f - c \cdot e \\ c \cdot d - a \cdot f \\ a \cdot e - b \cdot d \end{bmatrix} \hat{i} - \begin{bmatrix} a \cdot c \\ d \cdot f \\ a \cdot e - b \cdot d \end{bmatrix} \hat{j} + \begin{bmatrix} a \cdot c \\ d \cdot f \\ a \cdot e - b \cdot d \end{bmatrix} \hat{k}$$

$$= [bf - ce]\hat{i} - [af - cd]\hat{j} + [ac - bd]\hat{k}$$

$$= [bf - ce]\hat{i} + [cd - af]\hat{j} + [ac - bd]\hat{k}$$

$$= a[e\hat{i} \times \hat{j} + b\hat{j} \times \hat{i} + af\hat{i} \times \hat{k} + cd\hat{k} \times \hat{i} + bf\hat{j} \times \hat{k} + c \cdot k \times \hat{j}]$$

$$= [bf - ce]\hat{i} + [cd - af]\hat{j} + [ac - bd]\hat{k}$$

$$\text{or} = \begin{bmatrix} i & j & k \\ a & b & c \\ d & e & f \end{bmatrix}$$

Cross Product

Properties of $v \times w$

- $v \times w$ is perpendicular to both v & w
- $v \times w$ points in the direction according to the right-hand rule.
- $|v \times w| = |v||w|\sin\theta$ where θ is the positive angle between the vectors.

ex: $\langle 6, 0, 4 \rangle \times \langle -1, 1, 2 \rangle$ *↓ solve this*

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 0 & 4 \\ -1 & 1 & 2 \end{vmatrix} = \langle -4, -16, 6 \rangle$$

To check it with the original dot it.

$$\langle -4, -16, 6 \rangle \cdot \langle 6, 0, 4 \rangle = 0$$

$$\langle -4, -16, 6 \rangle \cdot \langle -1, 1, 2 \rangle = 0$$

Cross Products & Area

- If v & w define adjacent sides of a parallelogram, the area is $|v \times w|$.



$$|\vec{v} \times \vec{w}| = |v||w|\sin\theta$$

Lines:

Parametric Form

A line in $\mathbb{R}^n$ with pos P & dir v has parametric form

$$r(t) = P + tv$$

ex: Parametric form for

$$(-1, 0, -1) \text{ & } (1, 1, 1)$$

$$\vec{P} = \langle -1, 0, -1 \rangle, \vec{v} = \langle 2, 1, 2 \rangle$$

